

SHRI SHANKARACHARYA
TECHNICAL CAMPUS, BHILAI



श्री शंकराचार्य
टेक्निकल कैम्पस, भिलाई

Approved by AICTE, New Delhi
Constituent Unit of Shri Shankaracharya
Professional University, Bhilai

Accredited by NAAQ with "A" Grade
BTech courses" Accredited
by NBA, New Delhi

Best NSS Unit (President Award)
An ISO 9001:2015 Certified Institution

Roll No.:

Total Printed Pages: 2

AM100201

CT- I (March 26)

Subject- Applied Maths-II

B. TECH – II SEM (P2 - GROUP)

Maximum Marks: 40

Min Marks- 14

Duration: 2 hour

Note:-

- 1) All questions are compulsory.

Section 1

Q. (1) Objective Question:

(4x1=4)

1. The infinite series $\sum 1/n^2$ is convergent or divergent and why?
2. What is the necessary condition for an infinite series, $\sum u_n$ to be convergent?
3. State intermediate value property.
4. State D'moivre's theorem.

Or

Show that the Sequence:

$\frac{2^2}{2!}, \frac{3^2}{3!}, \frac{4^2}{4!}, \frac{5^2}{5!}, \dots$ is convergent.

Q. (2) Short Answer Type Question:

(3x4=12)

1. Test for convergence: $\sum (1 + \frac{1}{\sqrt{n}})^{-(n)^{\frac{1}{2}}}$.
2. Show that the equation; $x^7 - 3x^4 + 2x^3 - 1 = 0$ has at least four imaginary roots.
3. Separate: $\tan (x + i y)$ into real and imaginary parts.

Or

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Solve: $4x^4 - 20x^3 + 33x^2 - 20x + 4 = 0$.

Section 2

Q. (3) Descriptive Answer Type Question: (Solve any 3) (3x8=24)

1. Test the convergence of the series;

$$\sum (\sqrt{n^4 + 1} - \sqrt{n^4 - 1}).$$

2. Solve: $6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0$.

3. If $x_r = \cos \frac{\pi}{2r} + i \sin \frac{\pi}{2r}$,

Prove that: $x_1 x_2 \dots x_{\infty} = -1$.

Or

Test the convergence of the series;

$$1 + \frac{x}{2} + \frac{1.3}{2.4}x^2 + \frac{1.3.5}{2.4.6}x^3 + \frac{1.3.5.7}{2.4.6.8}x^4 + \dots$$



CLASS TEST - I (March 2026)

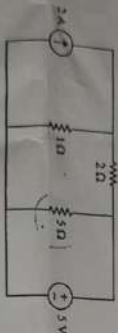
Subject Code	EE100107
Subject Name	Basic Electrical & Electronics Engineering

Q.1

Note :- Multiple choice question, carrying 1 marks each.
The type of circuit in which the elements are physically separable are called as:-

- Lumped circuit
- Linear circuit
- Distributed circuit
- Unilateral circuit

Q.2 The value of Thevenin's equivalent resistance R_{th} in the given figure is:-



Q.3

While applying Superposition theorem, what is done while removing the voltage source?

- They are replaced by an open circuit
- Their internal resistance are also removed
- They are replaced by a short circuit
- They are replaced by a short circuit and their internal resistances are left behind

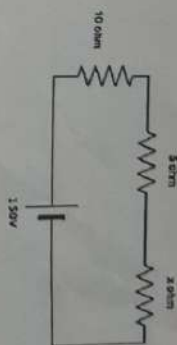
Q.4

- What is the value of x if the current in the circuit is 5A?
- 75 Ω
 - 65 Ω
 - 45 Ω
 - 45 Ω

Q.1

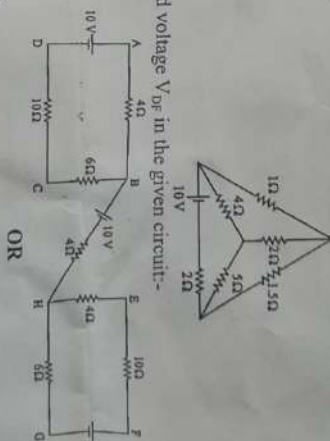
Note :- Each question carry 4 marks.
Find current in 4 Ω resistor of the given figure, using mesh analysis:-

Part - 2



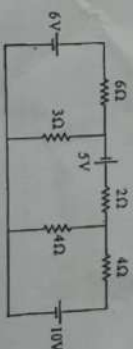
Q.2

Find voltage V_{df} in the given circuit:-



Q.3

Define:-
a) Active elements & Passive elements
b) Linear & Non linear circuits
Find current in 2 Ω resistor of the circuit in figure shown using Nodal method:-



OR
Find current in 6 Ω resistor of the given figure, using mesh analysis:-

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Section 2

Write Answer Type Questions: (Solve 4)

1. Find the value of x in the circuit.

$$V_1 = 10V, V_2 = 10V, V_3 = 10V$$

$$2x^2 + 31x^2 - 31x^2 + 32x - 6 = 0$$

$$+ 1.18 \frac{V}{m^2}$$

$$+ 1.18 \frac{V}{m^2}$$

$$+ 1.18 \frac{V}{m^2}$$

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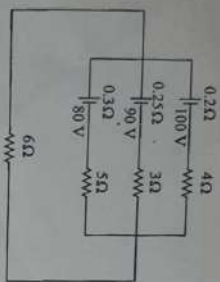
$$+ 1.18 \frac{V}{m^2}$$

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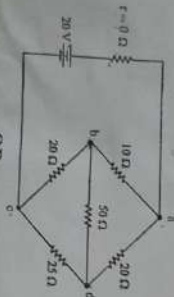
$$+ 1.18 \frac{V}{m^2}$$



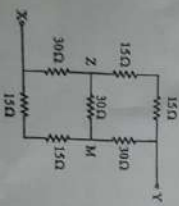
Part - 3

Q.1

Note :- Each question carry 8 marks.
Calculate V_{th} and R_{th} in 50Ω resistor of branch bd using Thevenin's Theorem:-



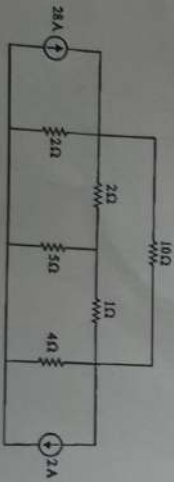
OR



Find the equivalent resistance between the terminals X and Y of the circuit shown:-

Q.2

Find the voltage across 10Ω resistor of the circuit in figure shown using Nodal method:-



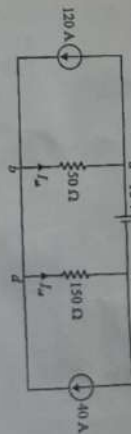
Q.3

Using Star Delta conversion find the resistance between terminals AB of the circuit shown in figure:-



OR

Find current through 50Ω resistor of branch ab by using Superposition Theorem:-





Class Test - I	Program B. Tech. 1 st Year		Semester : 2 nd March 2026		
Max. Time 2 Hours	SUBJECT: UHV-II		Code: HM100106	Max. Marks : 40	
CO 1:	To Understand and analyze the essential of human values and skills, self exploration, happiness				
CO 2:	To Evaluate coexistence of the Self with the body				
	Q. No	All questions are compulsory carrying 1 marks each	Marks	CO	PO
Objective Questions	1	What should be the human aspiration as per your syllabus? a. We want to live b. We want to live with fulfillment c. We want to live with continuous fulfillment d. All	1	1	1, 2, 3
	2	_____ on intention is the starting point for mutual development. a. Trust b. Belief c. Doubt d. None	1	1	1, 2, 4
Section-A	3	The human goal at the level of individual is a. Prosperity b. Fearlessness c. Co-existence d. Right understanding	1	2	1, 2, 4
	4	Right feeling within myself (I) is a. Definite b. Continuous c. Unconditional d. All	1	2	1, 2, 3
		All questions are compulsory carrying 4 marks each			
(Short answer Questions)	1	What is Value education and its needs in today's scenario?	4	1	1, 2, 3
	2	What is happiness and prosperity? Are they related to each other?	4	2	2, 3, 1
	3	Differentiate self and body on the basis of their needs.	4	3	2, 3, 4
Section-B (Long answer Questions)		Answer any three questions carrying 8 marks each (One question from each Unit)			
Unit -I	(a)	Define Self-exploration. Explain purposes and needs of self exploration.	8	1	1, 2, 3
	(b)	How do Relationship, Physical facilities and Right understanding help in fulfilling the human aspirations? Explain	8	1	2, 3, 4
Unit -2	(a)	Body is an instrument of the Self. Explain in details	8	2	2, 3, 4
	(b)	Define self and body. Explain the needs and responsibility of self towards the body.	8	2	1, 2, 3
Unit- 1 & 2	(a)	What are the basic guidelines of value education? Explain in brief	8	2	1, 3
	(b)	What do you mean by human and animal consciousness? Explain in brief	8	1	1, 2, 3

CT-1 B.Tech 2nd Semester 2025-26

Subject- Applied Physics
Maximum Marks – 40

Subject Code – AP100102
Duration – 2 hours

Note: All parts are compulsory in question 1 & 2. Attempt any one part in question 3, 4 & 5.

Multiple Choice Questions		
1.	a. The process of supplying energy to atoms to achieve population inversion is called: (a) Pumping (b) Cooling (c) Reflection (d) Absorption	1
	b. The wavelength of ruby laser light is approximately: (a) 632.8 nm (b) 694.3 nm (c) 1064 nm (d) 532 nm	1
	c. Optical fibre is used in medical field for: (a) Endoscopy (b) Cooking (c) Cleaning (d) Cooling	1
	d. One advantage of optical fibre communication is: (a) High signal loss (b) Large bandwidth (c) Heavy weight (d) Low data capacity	1
Short Answer Questions		
2.	a. Explain the terms population inversion, pumping and metastable state.	4
	b. Explain single-mode and multi-mode fibres.	4
	c. Explain dispersion in optical fibre.	4
Long Answer Questions		
3.	a. Explain the construction and working of a ruby laser with a labeled diagram. What is the drawback of ruby laser?	8
	b. (i) A laser beam has an intensity of $5 \times 10^3 \text{ W/m}^2$ and a cross-sectional area of 0.5 cm^2 . Calculate the power of the laser.	4
	(ii) A He-Ne laser emits light of 1 mW over a beam area of $2 \times 10^{-6} \text{ m}^2$. Calculate the intensity in W/m^2 .	4
4.	a. Explain construction and working of Helium - Neon laser? Why the discharge tube is taken narrow?	8
	b. An optical fibre has $n_1 = 1.48$, $n_2 = 1.46$, $a = 25 \mu\text{m}$, and $\lambda = 850 \text{ nm}$. Calculate: (a) NA (b) Acceptance angle (c) V-number	8
5.	a. Derive the expression for acceptance angle of an optical fibre in terms of the core and cladding refractive indices.	8
	b. (i) A step-index fibre has core radius $25 \mu\text{m}$ and $\text{NA} = 0.2$. Calculate the maximum acceptance angle in degrees.	4
	(ii) A step-index optical fibre has a core refractive index $n_1 = 1.50$ and cladding refractive index $n_2 = 1.48$. Calculate the numerical aperture (NA).	4

B.Tech. – Ist Year

Semester - 2

Session 2025-26

Class Test- 1

Subject Code: ME100105

Subject Name: Computer Aided Drafting & Design

Max. Marks: 40

Min. Marks: 14

Note: Sections A & B are compulsory. (Attempt any two questions from sections B.)

Attempt any one of the sections from C, D and E. (Attempt any two questions from C, D and E.)

Course Outcomes (CO). Upon successful completion, students will be able to:

CO1: Set up coordinate systems (HP, VP, RPP, LPP) and use basic drawing tools.

CO2 Create accurate front, top, and side views of points, lines, and planar shapes.

CO3: Generate orthographic projections and surface developments of regular

CO4. Convert orthographic views into correctly scaled isometric drawings.

CO5: Draft multidisciplinary layouts (parts with BOM, schematics, floor plans) and exchange models via standard file formats.

Q.no.	Questions	Marks	CO	BL	PI
SECTION-A					
1	1	Projection of a point in the second quadrant will be (a) Top view and front view above XY line (b) top view and front view below XY line (c) only front view appears below XY line (d) none			
		1	1	1,2	1,2
	2	In the first quadrant, the point is above H.P and In-front of V.P (a) True (b) False			
		1	1	1,2	1,2
	3	A point is 20 mm below HP and 30 mm behind VP. Its top view will be (a) 20 mm below XY (b) 20 mm below XY (c) 20 mm above XY (d) 30 mm above XY			
		1	1	1,2	1,2
	4	Projection of a point in the third quadrant will be (a) Top view in VP (b) Top view in HP (c) Top view in PP (d) None			
		1	1	1,2	1,2
SECTION-B Attempt any two questions					
2	1	What is the difference between first angle projection and third angle projection?			
		8	1	1,2	1,2

	2	What are the different line types used in engineering drawings?	8	1	1,2	1,2
	3	What are the principles and rules of dimensioning?	8	1	1,2	1,2
SECTION-C Attempt any two questions						
3	1	What is Projection of a Point? A point A is 30 mm above HP and 40 mm in front of VP. Draw its projections.	10	2	1,2	1,2
	2	What is the difference between first angle projection and third angle projection? Draw a projections of points:- A point A is 45 mm above HP and 50 mm in front of VP. Draw its projections.	10	2	1,2	1,2
	3	Explain the position of points after projection: 1. Point is above HP & Point is in front of VP 2. Point is above HP & behind VP 3. Point is below HP behind VP 4. Point is below HP & in front of VP	10	2	1,2	1,2

OR

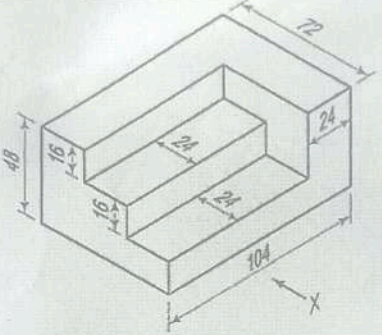
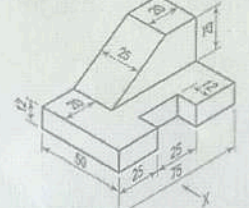
SECTION-D Attempt any two questions

4	1	Line AB is 75 mm long and it is inclined 30° to H.P. & parallel to V.P. End A is 12 mm above H.P. and 10 mm in front of V.P. Draw projections. Line is in 1st quadrant.	10	2	1.2	1.2
	2	Line AB is 60 mm long and it is inclined 45° to V.P. & parallel to H.P. End A is 15 mm above H.P. and 20 mm in front of V.P. Draw projections. Line is in 1st quadrant.	10	2	1.2	1.2
	3	Line AB is 60 mm long and it is 45° & 30° inclined to H.P. & V.P. respectively. End A is 20 mm above H.P. and 15 mm in front of V.P. Draw projections. Line is in 1st quadrant.	10	2	1.2	1.2

OR

SECTION-D Attempt any two questions

[illegible]

2		10	2	1,2,3	1,2
Draw the top view, front view by following first angle of projections.					
3		10	2	1,2,3	1,2